

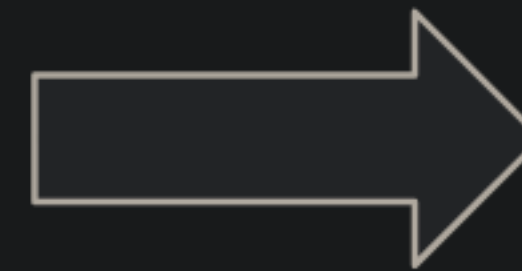
# Lesson 1 - The System Development Life Cycle (SDLC)



# Course Development Life Cycle

- Teaching slides
- Quizzes for self assessment, study and practice
- Labs to deliver practical lessons
- A feedback system
- ...

These are the **requirements for having a complete module.**



By writing down the requirements at **the right level of details** - not too much detail, not too abstract - I can work better



# Course Development Life Cycle

What am I doing here? I am doing a little bit of planning and analysis before jumping into making quiz questions or writing up activities. By writing down the requirements at the right amount of detail, not too much detail, not too abstract, I can work better.

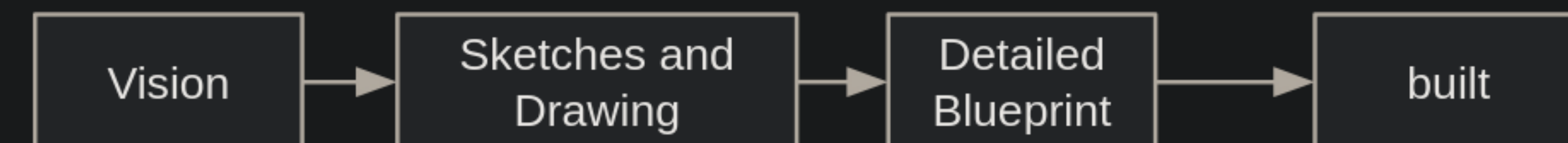
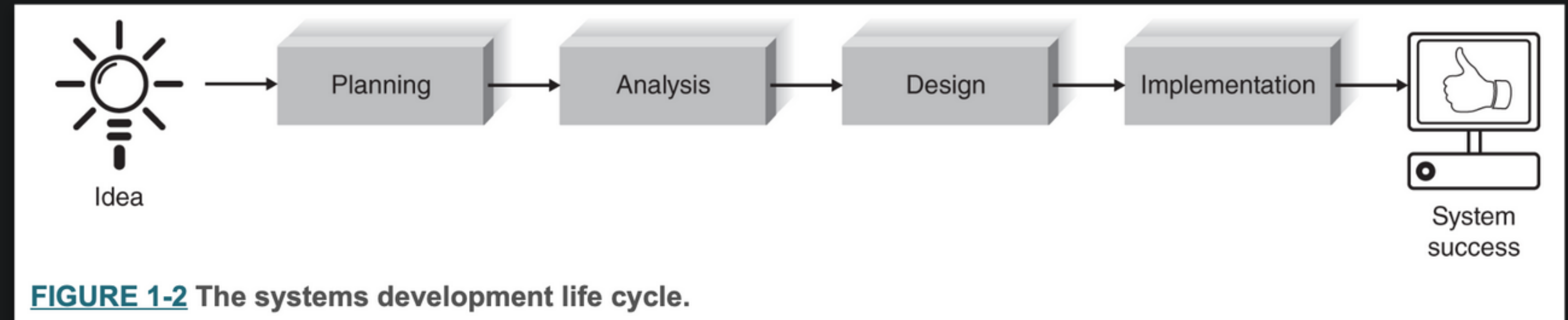
After writing down the top level requirements. I can focus on designing and building the teaching slides for module 1. Or import all chapter 1 questions to the question library on learning hub, etc. I can focus on building one module at a time, or build all the quizzes first since I have the library available to me. There are many ways to mix and match the tasks required to accomplish each of these requirements. But no matter how those tasks are mixed and match, building a course has a life cycle that includes planning, analyzing, designing and building.



# The System Development Life Cycle

Four fundamental phases:

1. Planning
2. Analysis
3. Design
4. Implementation





# The System Development Life Cycle

- SDLC is a process of **gradual refinement** → Each phase refines and elaborates on the work done previously
- **All projects** have elements of these four phases.
- **Important:** the project might go through the steps consecutively, incrementally, iteratively or in other **patterns**.



# Your Turn: Research Process Patterns



# Planning

| Phase                                                                       | Chapter | Step                       | Technique                                                                      | Deliverable                         |
|-----------------------------------------------------------------------------|---------|----------------------------|--------------------------------------------------------------------------------|-------------------------------------|
| <b>Planning</b>                                                             | 1       | Identify opportunity       | Project identification                                                         | System request                      |
| <u>Focus: Why build this system?</u>                                        | 1       | Analyze feasibility        | Technical feasibility<br>Economic feasibility<br>Organizational feasibility    | Feasibility study                   |
| How to structure the project?                                               | 2       | Develop project plan       | Scope management                                                               | Project plan                        |
| Primary outputs:<br>—System request with feasibility study<br>—Project plan |         |                            |                                                                                |                                     |
|                                                                             | 2       | Staff project              | Project staffing<br>Project charter                                            | —Staffing plan                      |
|                                                                             | 2       | Control and direct project | CASE repository<br>Standards<br>Documentation<br>Timeboxing<br>Risk management | —Standards list<br>—Risk assessment |



# Analysis

|                                                             |   |                                 |                                                                             |                          |
|-------------------------------------------------------------|---|---------------------------------|-----------------------------------------------------------------------------|--------------------------|
| <b>Analysis</b>                                             | 3 | Develop analysis strategy       | Business process automation                                                 | System proposal          |
| Focus: Who, what, where, and when for this system?<br><hr/> | 3 | Determine business requirements | Business process improvement<br>Business process reengineering<br>Interview | —Requirements definition |
| Primary output                                              |   |                                 | JAD session                                                                 |                          |
| —System proposal                                            |   |                                 | Questionnaire<br>Document analysis<br>Observation                           |                          |
|                                                             | 4 | Create use cases                | Use case analysis                                                           | —Use cases               |
|                                                             | 4 | Model processes                 | Data flow diagramming                                                       | —Process models          |
|                                                             | 5 | Model data                      | Entity relationship modeling<br>Normalization                               | —Data model              |



# Analysis

- A **concept** for the new system is created.
- Usually analysis begins by studying the current system (**as-is system**) and its **problems**, and then we envision a new system (**to-be system**).
- **Note:** Analysis phase actually produces a **high-level initial design**.



# Design

|                                          |    |                            |                                                                                                                   |                                                              |
|------------------------------------------|----|----------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Design</b>                            | 6  | Design physical system     | Design strategy                                                                                                   | Alternative matrix<br>System specification                   |
| <u>Focus: How will this system work?</u> | 7  | Design architecture        | Architecture design<br>Hardware and software selection                                                            | —Architecture report<br>—Hardware and software specification |
| Primary output:<br>—System specification | 8  | Design interface           | Use scenario<br>Interface structure<br>Interface standards<br>Interface prototype<br>Interface evaluation         | —Interface design                                            |
|                                          | 9  | Design programs            | Data flow diagramming<br>Program structure chart<br>Program specification                                         | —Physical process model<br>—Program design                   |
|                                          | 10 | Design databases and files | Data format selection<br>Entity relationship modeling<br>Denormalization<br>Performance tuning<br>Size estimation | —Database and file specification<br>—Physical data model     |



# Implementation

|                                                        |    |                                         |                                                               |                                                                     |
|--------------------------------------------------------|----|-----------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Implementation</b>                                  | 11 | Construct system                        | Programming                                                   | Test plan                                                           |
| <u>Focus: Delivery and support of completed system</u> |    | Software testing<br>Performance testing | Programs<br>Documentation                                     |                                                                     |
| Primary output:<br>—Installed system                   | 12 | Install system                          | Conversion strategy<br>selection                              | Migration plan<br>—Conversion plan<br>—Business contingency<br>plan |
|                                                        |    |                                         | Training                                                      | —Training plan                                                      |
|                                                        | 12 | Maintain system                         | Support selection<br>System maintenance<br>Project assessment | Support plan<br>Problem report<br>Change request                    |
|                                                        | 12 | Postimplementation                      | Postimplementation<br>audit                                   | Postimplementation audit<br>report                                  |



# Lesson 2 - Who is a Systems Analyst?



# The Systems Analyst

- They often serve as **change agents**
- They need to **work with all types of people** in the organization
- They need to understand **technology *and* business**



# Systems Analysts Skills

Six major categories:

1. Technical
2. Business
3. Analytical
4. Interpersonal
5. Management
6. Ethical



# Systems Analyst Roles

- Systems analyst
- Business analyst
- Requirements analyst
- Infrastructure analyst
- Software architect
- Change management analyst
- Project manager

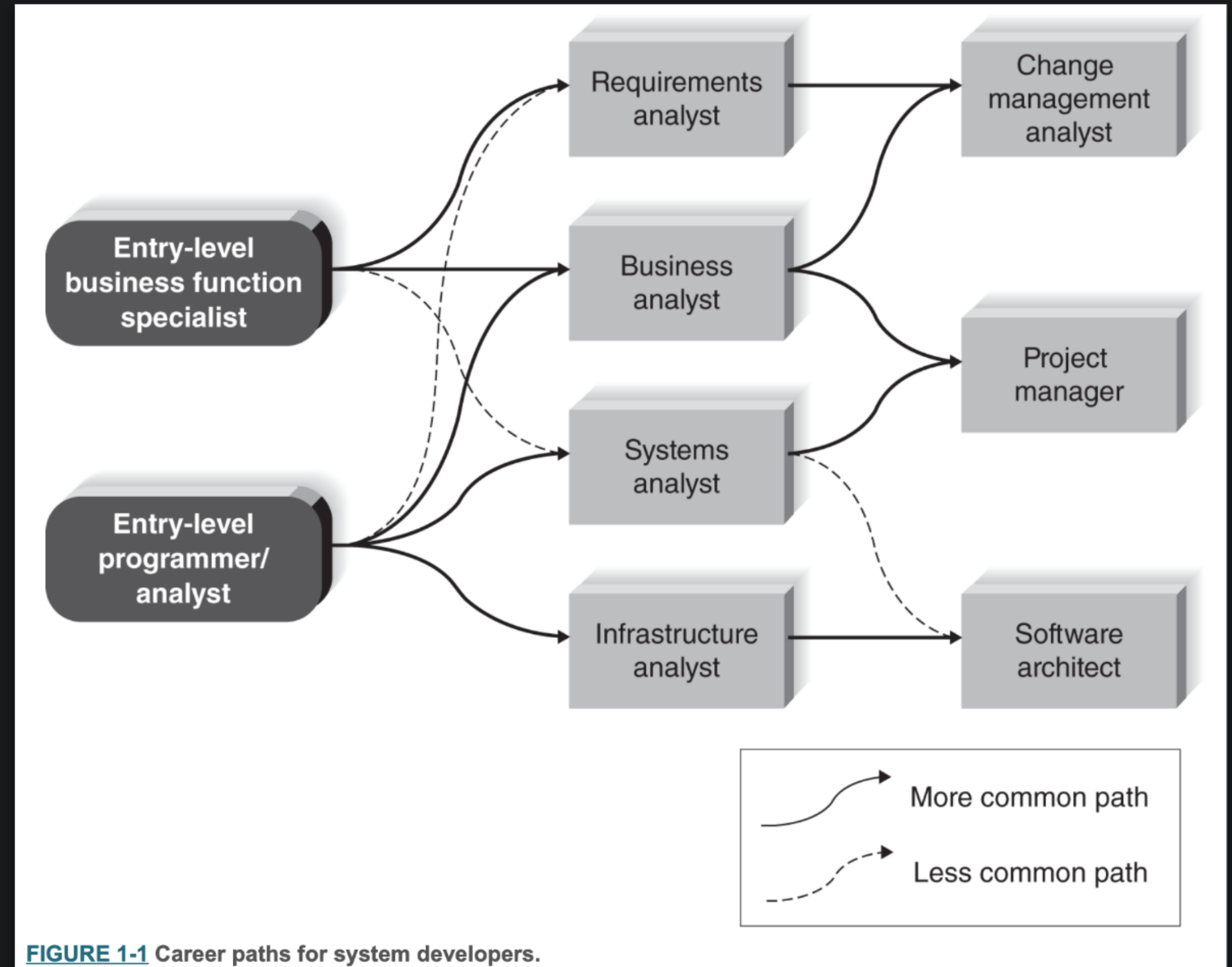


FIGURE 1-1 Career paths for system developers.



# Your Turn: Research Systems Analyst Job Postings

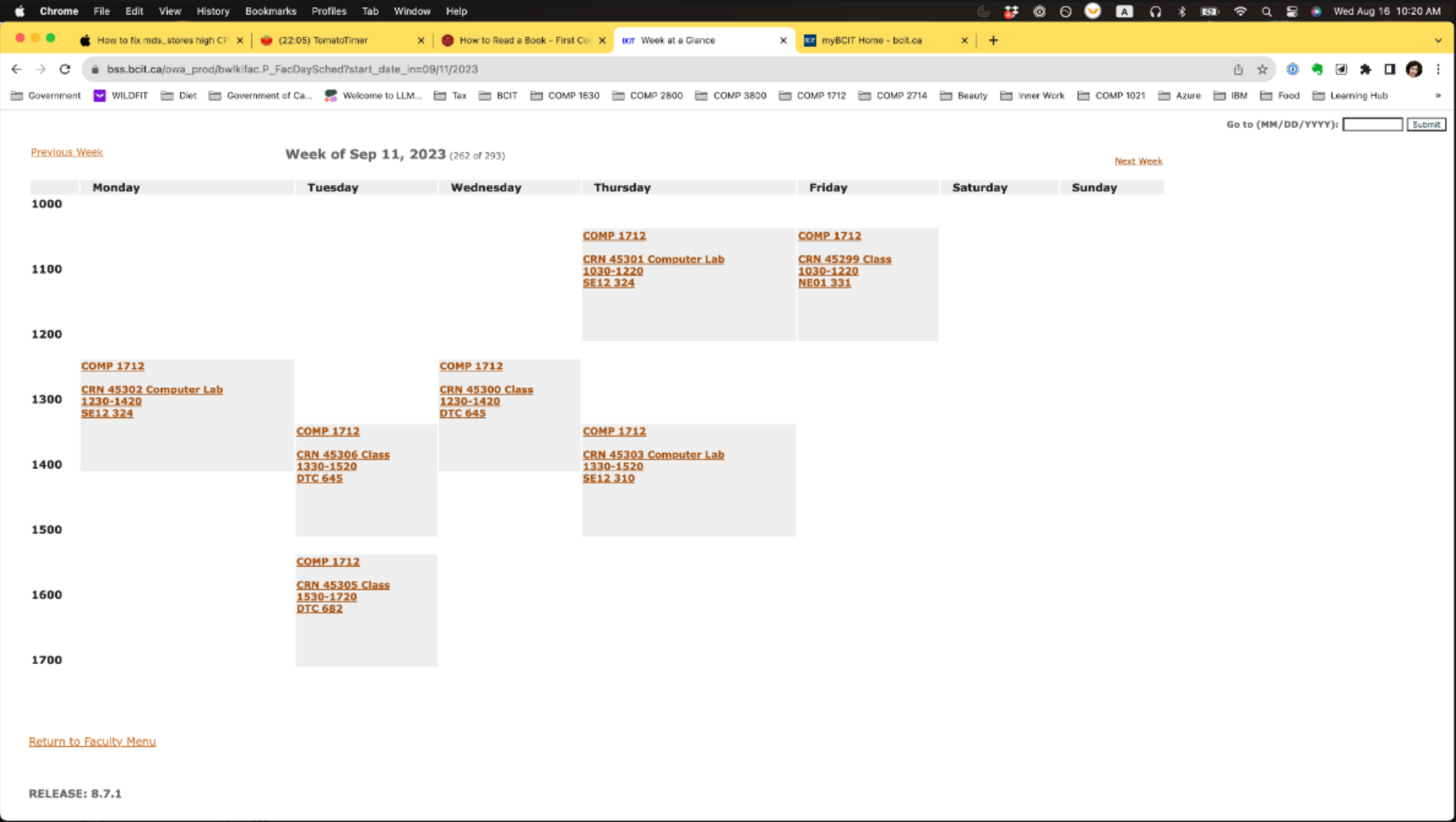


# Lesson 3 - How to identify and pitch a project?



# Faculty Schedules

As a faculty in BCIT I get to see my weekly schedule on my.bcit.ca. However I have to manually recreate that schedule on my personal calendar. If there was a way to export the schedule to the outlook calendar, that would save me a lot of time.





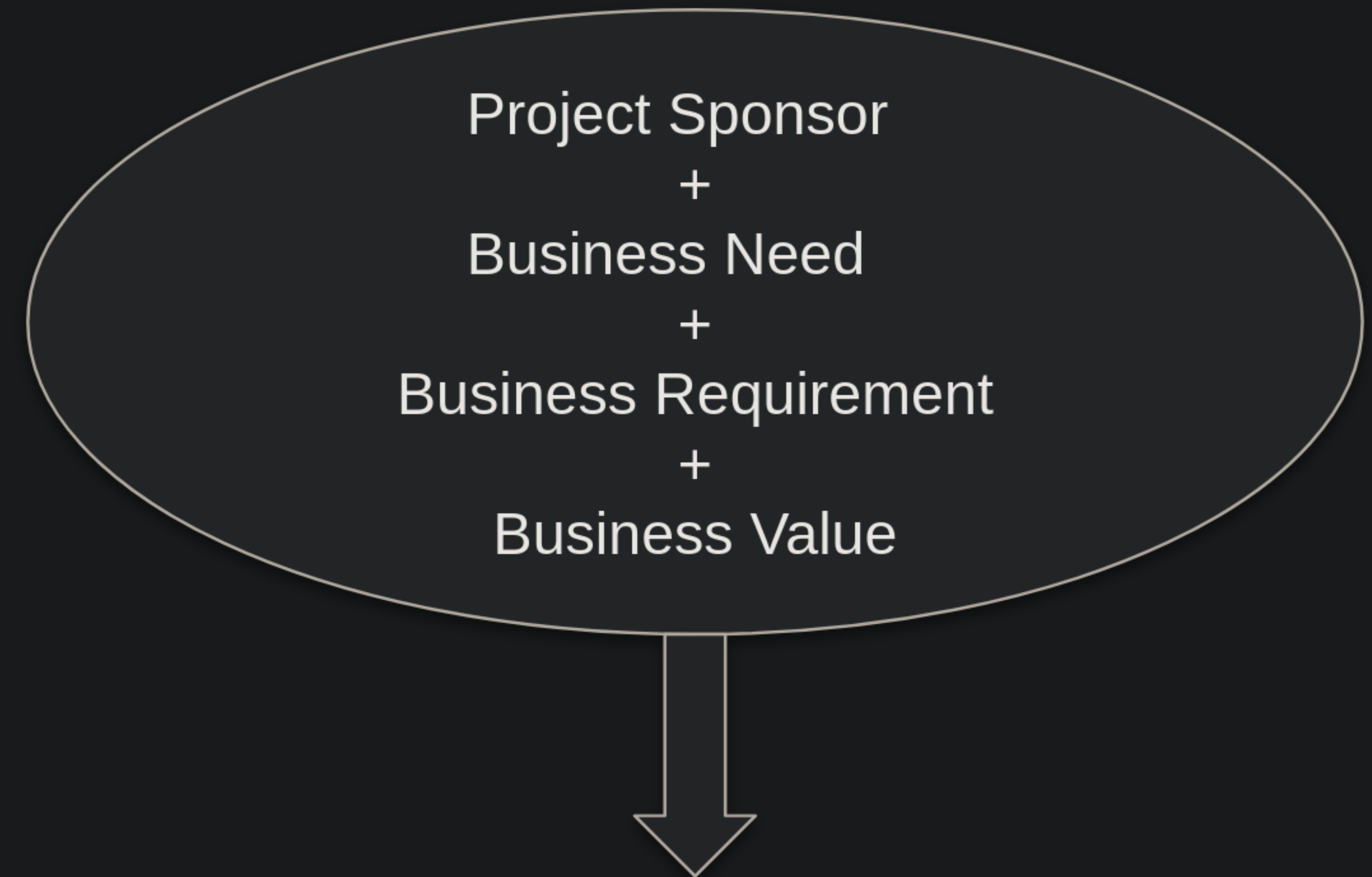
# Project Identification and Initiation

- Many of new projects grow out of **business process management (BPM)** initiatives.
- BPM is a methodology to **study and improve** business processes. It might lead to:
  - Business Process Automation (BPA)
  - Business Process Improvement (BPI)
  - Business Process Reengineering (BPR)



# Project Identification and Initiation

- When a strong **business need** is detected, a person or team who has an interest in the system's success, steps forward (**project sponsor**)
- **Business requirements** summarise the features the system must include so that the business needs are met.
  - Business requirements should be **explained at a high level** so the approval committee and project team understands the expectations.
- The project sponsor can also determine the **business value**, both **tangible and intangible** value



**System Request:** Formal Initiation of a project



# Your Turn: Read Project Identification and Initiation



# System Request

| Element                       | Description                                                                                                              | Examples                                                                                                                                                                                           |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Sponsor               | The person who initiates the project and who serves as the primary point of contact for the project on the business side | Several members of the finance department   Vice president of marketing   CIO   CEO                                                                                                                |
| Business Need                 | The business-related reason(s) for initiating the system                                                                 | Reach a new market segment   Offer a capability to keep up with competitors   Improve access to information   Decrease product defects   Streamline supply acquisition processes                   |
| Business Requirements         | The new or enhanced business capabilities that the system will provide                                                   | Provide online access to information   Capture customer demographic information   Include product search capabilities   Produce performance reports   Enhance online user support                  |
| Business Value                | The benefits that the system will create for the organization                                                            | 3% increase in sales   1% increase in market share   Reduction in headcount by 5 FTEs   \$200,000 cost savings from decreased supply costs   \$150,000 savings from removal of outdated technology |
| Special Issues or Constraints | Issues that pertain to the approval committee's decision                                                                 | Government-mandated deadline for May 30   System needed in time for the Christmas holiday season   Top-level security clearance needed by project team to work with data                           |
| FTE, full-time equivalent.    |                                                                                                                          |                                                                                                                                                                                                    |

**FIGURE 1-4** Elements of the system request form.



A committee reviews the system request and makes a decision. If approved the next step is to conduct a **feasibility study**.

|                | PLANNING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK CHECKLIST | <ul style="list-style-type: none"><li><input checked="" type="checkbox"/> Identify project.</li><li><input checked="" type="checkbox"/> Develop systems request.</li><li><input type="checkbox"/> Analyze technical feasibility.</li><li><input type="checkbox"/> Analyze economic feasibility.</li><li><input type="checkbox"/> Analyze organizational feasibility.</li><li><input type="checkbox"/> Perform project selection review.</li><li><input type="checkbox"/> Manage scope.</li><li><input type="checkbox"/> Staff project.</li><li><input type="checkbox"/> Create project charter.</li><li><input type="checkbox"/> Set up CASE repository.</li><li><input type="checkbox"/> Develop standards.</li><li><input type="checkbox"/> Begin documentation.</li><li><input type="checkbox"/> Assess and manage risk.</li></ul> |



# Your Turn: Applying the Concepts at DronTeq



# Lesson 4 - How to assess the feasibility of a project?



# Feasibility Analysis

- It is only wise to look at the project and see if we can actually do it (from a **technical, economic and organizational** point of view) before committing to the project.
- feasibility analysis is really about **analyzing the risks** involved in committing to a project.



# Technical Feasibility

- **Can we build it?**
- Technical risks we should consider:
  - **Familiarity with the business application area:** Less familiarity generates more risk.
    - New systems are riskier than extensions to existing systems
  - **Familiarity with technology:** Less familiarity generates more risk.
  - **Project size:** Large projects have more risk.
    - Project size might be measured in various ways: e.g. #of people involved, time required, # of features, etc.
  - **Compatibility:** The harder it is to integrate the system with the company's existing technology, the higher the risk will be.
    - Systems are barely built in a vacuum.



# Technical Feasibility

To assess the technical feasibility:

- Compare project to **prior projects**.
- Consult with experienced internal/external **professionals**



# Economic Feasibility

- **Should we build the system?**
- This is about **financial worthiness** of the project.
- Also called **cost-benefit analysis**
- We should consider:
  - Development costs
  - Annual operating costs
  - Annual benefits (cost savings and/or increased revenues)
  - Intangible benefits and costs
- **Cash flows** (both inflow and outflow) are estimated over some future period.
- Then these flows are **evaluated** using several techniques.
- **Goal:** to judge whether the projected benefits justify the costs.



# Example: Cash Flow Projections

|                                                  | Year 0    | Year 1   | Year 2   | Year 3 | Total   |
|--------------------------------------------------|-----------|----------|----------|--------|---------|
| Total Benefits                                   |           | 45,000   | 50,000   | 57,000 | 152,000 |
| Total Costs                                      | 100,000   | 10,000   | 12,000   | 16,000 | 138,000 |
| ③ Net Benefits<br>(Total Benefits – Total Costs) | (100,000) | 35,000   | 38,000   | 41,000 | 14,000  |
| ④ Cumulative Net Cash Flow                       | (100,000) | (65,000) | (27,000) | 14,000 |         |

**FIGURE 1-8** Simple cash flow projection.



The value of the  
project must be  
measured over time.



# Return on Investment (ROI)

- Measures the **average rate of return**. (Why is it called rate?)
- ROI is commonly used in practice, but it is hard to interpret and **should not be used as the only measure of a project's worth**.

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$
$$\text{ROI} = \frac{152,000 - 138,000}{138,000} = \frac{14,000}{138,000} = 10.14\%$$



# Break-Even Point (BEP)

- Another measure for a project's worth.
- Also called the **payback method**
- Measures the number of years it takes to recover the original investment.
  - That's when the *cumulative cash flow becomes positive*.

$$BEP = \text{Number of years of negative cash flow} + \frac{\text{That Year's Net Cash Flow} - \text{That Year's Cumulative Cash Flow}}{\text{That Year's Net Cash Flow}}$$

$$BEP = 2 + \frac{41,000 - 14,000}{41,000} = 2.68 \text{ Years}$$



# Discounted Cash Flow Technique

- ROI and BEP techniques both have a weakness:
  - They do not recognize the **time value of money**. The timing of cash flows is ignored.
- Is a dollar in year 3 the same as a dollar right now?!
- **Time makes money grow.**
- A dollar in the future is worth less than a dollar today.
- The **present value** of a future dollar is the amount of money that, when invested today at a specified **rate of return**, grows to exactly one dollar at a certain point in the future.

$$PV = \frac{\text{Cash flow amount}}{(1 + \text{Rate of return})^n}$$



# Rate of Return

- Sometimes called the *required* rate of return or the **cost of obtaining the capital** needed to fund the project.
- Many organizations determine the appropriate rate of return to analyze IT investments.

|                      | Year 0  | Year 1 | Year 2 | Year 3 | Total   |
|----------------------|---------|--------|--------|--------|---------|
| Total Benefits       |         | 45,000 | 50,000 | 57,000 |         |
| PV of Total Benefits |         | 40,909 | 41,322 | 42,825 | 125,056 |
| Total Costs          | 100,000 | 10,000 | 12,000 | 16,000 |         |
| PV of Total Costs    | 100,000 | 9,091  | 9,917  | 12,021 | 131,029 |

**FIGURE 1-9** Discounted cash flow projection.



# Net Present Value (NPV)

$$\begin{aligned}\text{NPV} &= \sum \text{PV of Total Benefits} - \sum \text{PV of Total Costs} \\ &= \$125,056 - \$131,029 = (\$5,973)\end{aligned}$$

- $\text{NPV} > 0 \rightarrow$  Project is economically acceptable
- $\text{NPV} < 0 \rightarrow$  For the required rate of return the project should not be accepted.
- **Notice** that while using the naive techniques of ROI and BEP the project might appear acceptable, the NPV reveals that for the required rate of return, the project is not feasible.



|                                  | Year 0    | Year 1   | Year 2   | Year 3 | Total   |
|----------------------------------|-----------|----------|----------|--------|---------|
| Total Benefits                   |           | 45,000   | 50,000   | 57,000 | 152,000 |
| Total Costs                      | 100,000   | 10,000   | 12,000   | 16,000 | 138,000 |
| Net Benefits                     |           |          |          |        |         |
| ③ (Total Benefits – Total Costs) | (100,000) | 35,000   | 38,000   | 41,000 | 14,000  |
| ④ Cumulative Net Cash Flow       | (100,000) | (65,000) | (27,000) | 14,000 |         |

**FIGURE 1-8** Simple cash flow projection.

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

$$\text{ROI} = \frac{152,000 - 138,000}{138,000} = \frac{14,000}{138,000} = 10.14\%$$

$$\text{BEP} = \text{Number of years of negative cash flow} + \frac{\text{That Year's Net Cash Flow} - \text{That Year's Cumulative Cash Flow}}{\text{That Year's Net Cash Flow}}$$

$$\text{BEP} = 2 + \frac{41,000 - 14,000}{41,000} = 2.68 \text{ Years}$$

$$\begin{aligned} \text{NPV} &= \sum \text{PV of Total Benefits} - \sum \text{PV of Total Costs} \\ &= \$125,056 - \$131,029 = (\$5,973) \end{aligned}$$



# Steps to conduct an economic feasibility analysis.

|                                               |                                                                                                                                                                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Identify Costs and Benefits</b>         | List the tangible costs and benefits for the project. Include both one-time and recurring costs.                                                                                                                                                            |
| <b>2. Assign Values to Costs and Benefits</b> | Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if possible.                                                                                                          |
| <b>3. Determine Cash Flow</b>                 | Forecast what the costs and benefits will be over a certain period, usually, 3–5 years. Apply a growth rate to the values, if necessary.                                                                                                                    |
| <b>4. Assess Project's Economic Value</b>     | Evaluate the project's expected returns in comparison to its costs. Use one or more of the following evaluation techniques:                                                                                                                                 |
| • <b>Return on Investment (ROI)</b>           | Calculate the rate of return earned on the money invested in the project, using the ROI formula.                                                                                                                                                            |
| • <b>Break-Even Point (BEP)</b>               | Find the year in which the cumulative project benefits exceed cumulative project costs. Apply the break-even formula, using figures for that year. This calculation measures how long it will take for the system to produce benefits that cover its costs. |
| • <b>Net Present Value (NPV)</b>              | Restate all costs and benefits in today's dollar terms (present value), using an appropriate discount rate. Determine whether the total present value of benefits is greater than or less than the total present value of costs.                            |

**FIGURE 1-10** Steps to conduct an economic feasibility analysis.



# Example of costs and benefits for economic feasibility

| Development Costs                                                                                                                                                           | Operational Costs                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Development team salaries<br>Consultant fees<br>Development training<br>Hardware and software<br>Vendor installation<br>Office space and equipment<br>Data conversion costs | Software upgrades<br>Software licensing fees<br>Hardware repair and upgrades<br>Cloud storage fees<br>Operational team salaries<br>Communications charges<br>User training |
| Tangible Benefits                                                                                                                                                           | Intangible Benefits                                                                                                                                                        |
| Increased sales<br>Reductions in staff<br>Reductions in inventory<br>Reductions in IT costs<br>Better supplier prices                                                       | Increased market share<br>Increased brand recognition<br>Higher-quality products<br>Improved customer service<br>Better supplier relations                                 |

**FIGURE 1-11** Example of costs and benefits for economic feasibility.



# Organizational Feasibility

- You need a good understanding of the stakeholders (**Stakeholder Analysis**)
  - **Stakeholder:** a person, group or organization that can affect (or be affected by) a new system.
- You also need to understand the business objectives well in order to see if the project aligns with where the business wants to go. (**Strategic Alignment**)
- The goal is to assess how well the system ultimately will be accepted by its users and incorporated into the ongoing operations of the organizations.



|                           | Role                                                                                                                                                                                                                                                        | To Enhance Organizational Feasibility                                                                                                                                                                                                                                                                                                                            |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Champion                  | A champion: <ul style="list-style-type: none"> <li>• Initiates the project</li> <li>• Promotes the project</li> <li>• Allocates his or her time to the project</li> <li>• Provides resources</li> </ul>                                                     | <ul style="list-style-type: none"> <li>• Make a presentation about the objectives of the project and the proposed benefits to those executives who will benefit directly from the system.</li> <li>• Create a prototype of the system to demonstrate its potential value.</li> </ul>                                                                             |
| Organizational Management | Organizational managers: <ul style="list-style-type: none"> <li>• Know about the project</li> <li>• Budget enough money for the project</li> <li>• Encourage users to accept and use the system</li> </ul>                                                  | <ul style="list-style-type: none"> <li>• Make a presentation to the management about the objectives of the project and the proposed benefits.</li> <li>• Market the benefits of the system, using memos and organizational newsletters.</li> <li>• Encourage the champion to talk about the project with his organizational newsletters or her peers.</li> </ul> |
| System Users              | Users: <ul style="list-style-type: none"> <li>• Make decisions that influence the project</li> <li>• Perform hands-on activities for the project</li> <li>• Ultimately determine whether the project is successful using or not using the system</li> </ul> | <ul style="list-style-type: none"> <li>• Assign users official roles on the project team.</li> <li>• Assign users specific tasks to perform, with clear deadlines.</li> <li>• Ask for feedback from users regularly (e.g., at weekly meetings).</li> </ul>                                                                                                       |

**FIGURE 1-14** Important stakeholders for organizational feasibility.



# Your Turn: Lab Preparation